



BAULKHAM HILLS HIGH SCHOOL

2017
YEAR 12 HALF YEARLY
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

Total marks – 70

Section I Pages 2 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II Pages 6 – 10

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

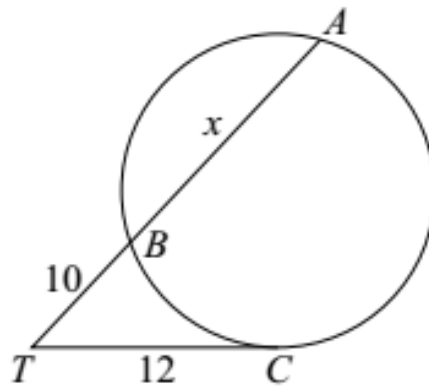
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

- 1 The point P divides the interval from $A(-1,-2)$ to $B(5,1)$ internally in the ratio 2:1. What are the coordinates of P ?
- (A) $\left(0, -\frac{3}{2}\right)$
- (B) $(1,-1)$
- (C) $\left(2, -\frac{1}{2}\right)$
- (D) $(3,0)$
- 2 When $2x^3 - 3x^2 + 2a - 4$ is divided by $x - 1$ the remainder is -5 . What is the value of a ?
- (A) 2
- (B) 0
- (C) -2
- (D) -3
- 3 If $\cos x = \frac{3}{4}$ and $\sin x < 0$, which of the following is the exact value of $\sin 2x$?
- (A) $-\frac{3\sqrt{7}}{8}$
- (B) $\frac{\sqrt{7}}{4}$
- (C) $-\frac{\sqrt{7}}{4}$
- (D) $\frac{3\sqrt{7}}{4}$

- 4 Which of the following is a point on the parabola $x^2 = 4ay$?
- (A) $(0, a)$
- (B) $(0, -a)$
- (C) $\left(\frac{2a}{r}, \frac{a}{r^2}\right)$
- (D) $(aq^2, 2aq)$
- 5 How many ways can 8 people be arranged around a circular table if Dineth must sit between Zhan and Xianyi?
- (A) 120
- (B) 240
- (C) 720
- (D) 1440

6



In the diagram above, the tangent at C meets the secant AB at T . Given that $AB = x$, $BT = 10$ and $CT = 12$, the value of x is:

- (A) 2
- (B) $4\frac{2}{5}$
- (C) 8
- (D) $14\frac{2}{5}$

7 $\lim_{x \rightarrow 0} \frac{\tan 3x}{2x}$ is equal to:

(A) 0

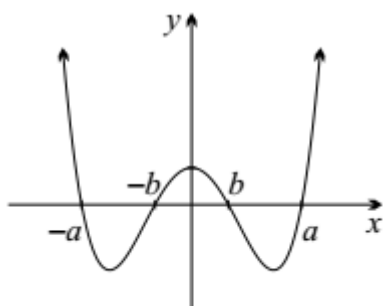
(B) $\frac{2}{3}$

(C) 1

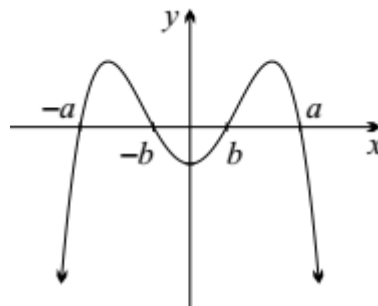
(D) $\frac{3}{2}$

8 Which diagram best represents $y = (x - a)^2(b^2 - x^2)$, where $a > b$?

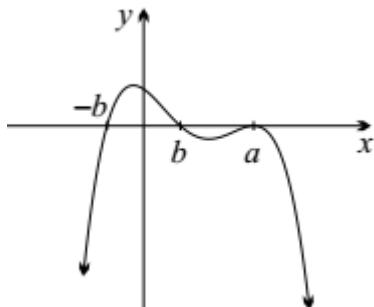
(A)



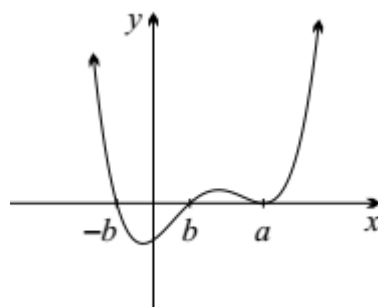
(B)



(C)



(D)



9 What is the value of $\cos^{-1}[\cos(3\pi + \alpha)]$ where α is an acute angle?

(A) α

(B) $\pi - \alpha$

(C) $\pi + \alpha$

(D) $3\pi + \alpha$

- 10 Given that the roots of $x^2 - 2x - 1 = 0$ are $\tan \alpha$ and $\tan \beta$, what is the value of $\alpha + \beta$?

(A) $\frac{\pi}{4}$

(B) $-\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) $-\frac{\pi}{2}$

END OF SECTION I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour 45 minutes for this section

Answer each question on the appropriate answer sheet. Each answer sheet must show your NES A#. Extra paper is available.

In Questions 11 to 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a *separate* answer sheet

Marks

(a) Prove that $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} = 2$ 2

(b) Differentiate $\tan^{-1} \frac{5x}{4}$ 2

(c) Find

(i) $\int \cos^2 x \, dx$ 2

(ii) $\int \frac{dx}{\sqrt{1-2x^2}}$ 2

(d) Express $8\cos x + 15\sin x$ in the form $R\cos(x - \alpha)$, giving α correct to the nearest degree. 2

(e) Find the general solution to $2\sin x = \sqrt{3}$ 2

(f) (i) How many nine letter arrangements can be made using the letters of the word; 1

SCHOOLIES

(ii) In how many of the arrangements in part (i) do the vowels appear together? 1

(iii) In how many of the arrangements in part (i) does the word **COOL** appear? 1

Question 12 (15 marks) Use a *separate* answer sheet

(a) Find the greatest value of $\frac{\cos^2\left(\frac{\pi}{2} - \theta\right) - 2\cos^2\theta}{24}$ for $0 \leq \theta \leq \frac{\pi}{2}$ 3

(b) (i) Solve the inequality $\frac{x-5}{x-1} \leq -1$ 3.

(ii) Hence, or otherwise, solve $\frac{\cos\alpha - 4}{\cos\alpha} \leq -1$, for $0 \leq \alpha \leq \pi$ 2

(c) In January 1995 the purebred dingo population on Fraser Island was 300. The population, P , since then can be modelled by;

$$P = 80 + Ae^{kt}$$

where A and k are constants, and t is the time since January 1995, in years.

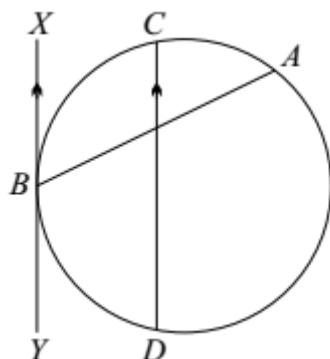
(i) Show that this model is a solution to the differential equation 1

$$\frac{dP}{dt} = k(P - 80)$$

(ii) In January 2015 it was found that the purebred population had dropped to 162. Show that purebred dingo population is decreasing at annual rate of approximately 5% per year. 2

(iii) Assuming this pattern continues, what will the purebred dingo population be in January 2050? 1

(d) 3



In the diagram, AB and CD are intersecting chords. The tangent at B is parallel to CD .

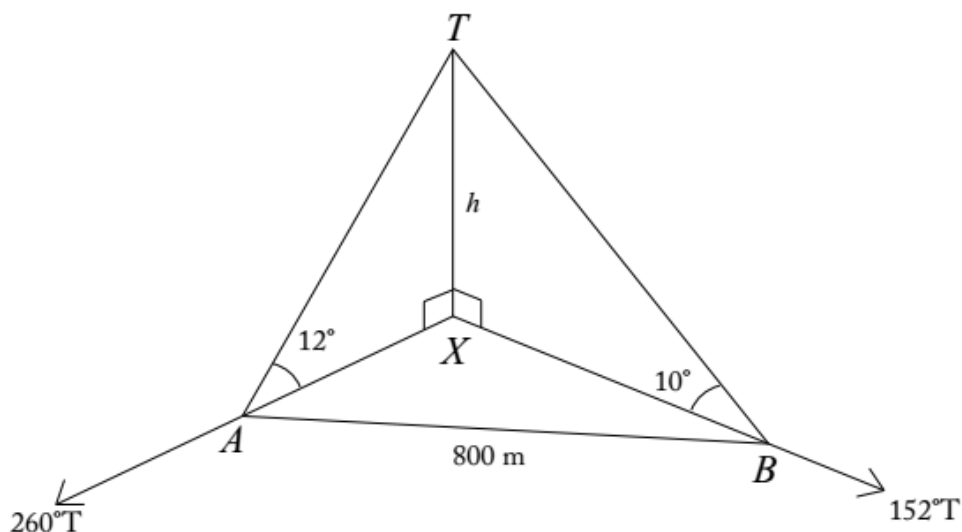
Copy this diagram into your answer booklet and prove that AB bisects $\angle CAD$

Question 13 (15 marks) Use a *separate* answer sheet

- (a) The radius of a circle is increasing such that the rate of increase of the area is $\pi^2 r \text{ cm}^2/\text{s}$. 2

Calculate the rate of increase of the radius.

(b)



In the diagram above; TX represents a vertical tower of height h metres standing on the horizontal plane AXB .

Rachel and Marina are standing 800 metres apart on the same plane. Rachel is at point A on a bearing of $260^\circ T$ from the tower and the angle of elevation to the top of the tower is 12° . Marina is at point B on a bearing of $152^\circ T$ from the tower and the angle of elevation to the top of the tower is 10° .

- (i) Explain why $\angle AXB = 108^\circ$ 1
- (ii) Express AX in terms of h 1
- (iii) Find the height of the tower to the nearest metre 2
- (c) Use mathematical induction to prove that; 3

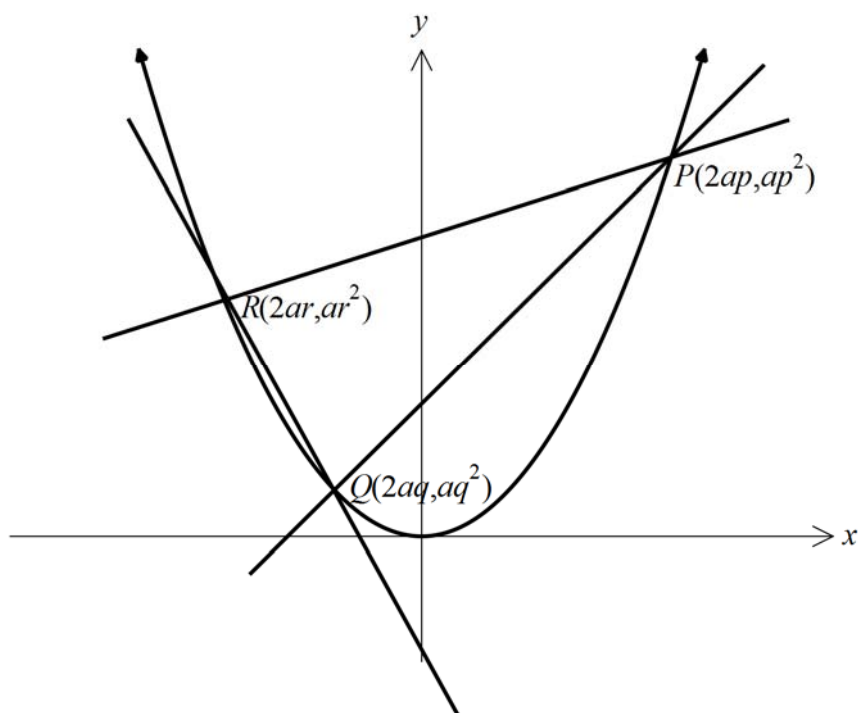
$$\sum_{r=1}^n \frac{5-4r}{5^r} = \frac{n}{5^n}$$

Question 13 continues on page 9

Marks

Question 13 (continued)

- (d) The diagram shows the parabola $x^2 = 4ay$. The distinct points $P(2ap, ap^2)$, $Q(2aq, aq^2)$ and $R(2ar, ar^2)$ lie on the parabola such that the normal to the parabola at Q and R both pass through the point P .



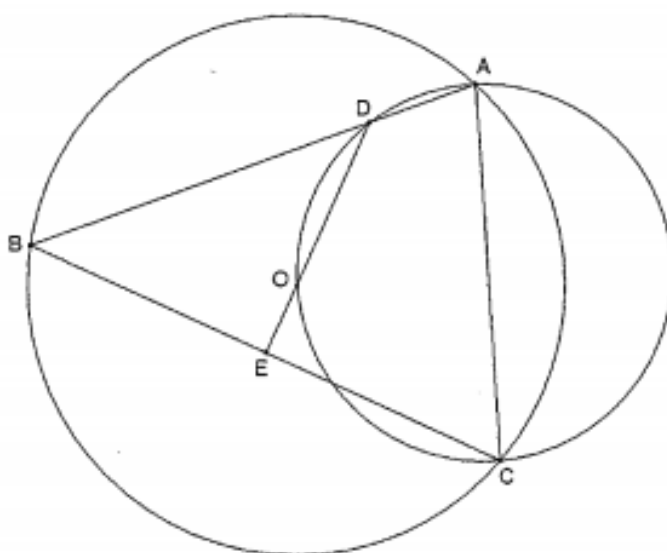
- | | | |
|-------|--|---|
| (i) | Given that the equation of the normal at Q is $x + qy = aq^3 + 2aq$, show that $q^2 + pq + 2 = 0$ | 2 |
| (ii) | Show that the equation of the chord QR is given by $(q + r)x - 2y = 2aqr$ | 2 |
| (iii) | Show that QR always passes through the point $(0, -2a)$ | 2 |

End of Question 13

Question 14 (15 marks) Use a *separate* answer sheet

- (a) (i) State the domain and range of $y = 2\cos^{-1}(1-x)$ 2
- (ii) Sketch $y = 2\cos^{-1}(1-x)$ 1
- (iii) On the same set of axes as part (ii), sketch $y = -\pi x + 2\pi$ 1
- (iv) Explain why $\int_0^2 2\cos^{-1}(1-x) dx = \int_0^2 (-\pi x + 2\pi) dx$ 1
- (v) Without integrating, evaluate $\int_0^2 2\cos^{-1}(1-x) dx$ 1

- (b) ABC is a triangle inscribed in a circle with centre O . A second circle through the points A, C, O cuts AB at D . DO is produced to meet BC at E .

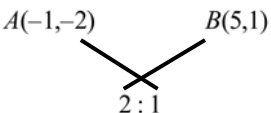
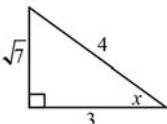


Copy the diagram into your answer booklet

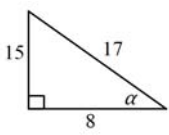
- (i) Prove that $\angle BOE = \angle BAC$ 3
- (ii) Prove that $BE = CE$ 2
- (c) (i) Use the factor theorem to show that $(a + b - c)$ is a factor of 2
- $$(a + b + c)^3 - 6(a + b + c)(a^2 + b^2 + c^2) + 8(a^3 + b^3 + c^3)$$
- (ii) Hence factorise $(a + b + c)^3 - 6(a + b + c)(a^2 + b^2 + c^2) + 8(a^3 + b^3 + c^3)$ 2

End of paper

BAULKHAM HILLS HIGH SCHOOL
YEAR 12 HALF YEARLY EXAMINATION 2017 SOLUTIONS

Solution		Marks	Comments
SECTION I			
1. D –	 $P = \left(\frac{1 \times -1 + 2 \times 5}{2 + 1}, \frac{1 \times -2 + 2 \times 1}{2 + 1} \right)$ $= \left(\frac{9}{3}, \frac{0}{3} \right)$ $= (3, 0)$	1	
2. B –	$P(x) = 2x^3 - 3x^2 + 2a - 4$ $P(1) = -5$ $2 - 3 + 2a - 4 = -5$ $2a = 0$ $a = 0$	1	
3. A –	$\sin 2x = 2 \sin x \cos x$ $= 2 \times -\frac{\sqrt{7}}{4} \times \frac{3}{4}$ $= -\frac{3\sqrt{7}}{8}$ 	1	
4. C –	$x^2 = \left(\frac{2a}{r} \right)^2$ $= \frac{4a^2}{r^2}$ $4ay = 4a \times \frac{a}{r^2}$ $= \frac{4a^2}{r^2}$ $= x^2$	1	
5. B –	$\text{Ways} = 2! \times 5!$ $= 240$ Zhan and Xianyi must sit either side of Dineth = 2! Arrange group of three plus five other people Arrange 6 objects in a circle = 5!	1	
6. B –	$AT \times BT = CT^2$ (square of tangent equals product of intercepts) $10(x + 10) = 12^2$ $10x + 100 = 144$ $10x = 44$ $x = \frac{22}{5} = 4\frac{2}{5}$	1	
7. D –	$\lim_{x \rightarrow 0} \frac{\tan 3x}{2x} = \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{3}{2 \cos 3x}$ $= \frac{3}{2}$	1	
8. C –	$(x - a)^2 \Rightarrow$ double root at $x = a$ $(b^2 - x^2) = (b - x)(b + x) \Rightarrow$ single roots at $x = \pm b$ as $x \rightarrow -\infty$ y behaves like leading term, $-x^3$	1	
9. B –	$\cos(3\pi + \alpha) = \cos(\pi + \alpha)$ $= -\cos \alpha$ $\cos^{-1}[\cos(3\pi + \alpha)] = \cos^{-1}(-\cos \alpha)$ $= \pi - \cos^{-1} \cos \alpha$ $= \pi - \alpha$	1	
10. A –	$\tan \alpha + \tan \beta = 2$ & $\tan \alpha \tan \beta = -1$ $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$ $= \frac{2}{1 + 1}$ $= 1$ $\alpha + \beta = \frac{\pi}{4}$	1	

SECTION II

Solution		Marks	Comments
QUESTION 11			
11(a)	$\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} = \frac{\sin 3\theta \cos \theta - \cos 3\theta \sin \theta}{\sin \theta \cos \theta}$ $= \frac{\sin(3\theta - \theta)}{\sin \theta \cos \theta}$ $= \frac{1}{2} \times 2 \sin \theta \cos \theta$ $= \frac{2 \sin 2\theta}{\sin 2\theta}$ $= 2$	2	2 marks • Correct solution 1 mark • Correctly uses $\sin 2\theta$ result
11(b)	$\frac{d}{dx} \left(\tan^{-1} \frac{5x}{4} \right) = \frac{\frac{5}{4}}{1 + \frac{25x^2}{16}}$ $= \frac{20}{16 + 25x^2}$ OR $\frac{d}{dx} \left(\tan^{-1} \frac{5x}{4} \right) = \frac{d}{dx} \left(\tan^{-1} \frac{x}{\frac{4}{5}} \right)$ $= \frac{\frac{4}{5}}{\frac{16}{25} + x^2}$ $= \frac{20}{16 + 25x^2}$	2	2 marks • Correct solution 1 mark • Obtains a denominator of $16 + 25x^2$, or equivalent
11(c) (i)	$\int \cos^2 x \, dx = \frac{1}{2} \int (1 + \cos 2x) \, dx$ $= \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) + c$	2	2 marks • Correct solution 1 mark • Correctly uses $\cos 2\theta$ result
11(c) (ii)	$\int \frac{dx}{\sqrt{1-2x^2}} = \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\frac{1}{2}-x^2}}$ $= \frac{1}{\sqrt{2}} \sin^{-1} \left(\frac{x}{\frac{1}{\sqrt{2}}} \right) + c$ $= \frac{1}{\sqrt{2}} \sin^{-1} \sqrt{2}x + c$	2	2 marks • Correct solution 1 mark • Uses correct standard integral
11(d)	 $\alpha = \tan^{-1} \frac{15}{8}$ $= 61.9275...^\circ$ $8\cos x + 15\sin x = 17\cos(x - 62^\circ)$	2	2 marks • Correct solution 1 mark • Correctly finds R or α
11(e)	$2\sin x = \frac{\sqrt{3}}{2}$ $\sin x = \frac{\sqrt{3}}{4}$ $x = \pi k + (-1)^k \sin^{-1} \frac{\sqrt{3}}{4}$ $x = \pi k + (-1)^k \left(\frac{\pi}{6} \right) \text{ where } k \text{ is an integer}$	2	2 marks • Correct solution 1 mark • establishes $\frac{\pi}{6}$ as the principal angle • uses the correct general angle formula
11(f) (i)	$\text{Ways} = \frac{9!}{2!2!}$ $= 90720$	1	1 mark • Answer may be left in factorial notation.
11(f) (ii)	$\text{Ways} = \frac{4!}{2!} \times \frac{6!}{2!}$ $= 4320$	1	1 mark • Answer may be left in factorial notation.
11(f) (iii)	$\text{Ways} = 1 \times \frac{6!}{2!}$ $= 360$	1	1 mark • Answer may be left in factorial notation.

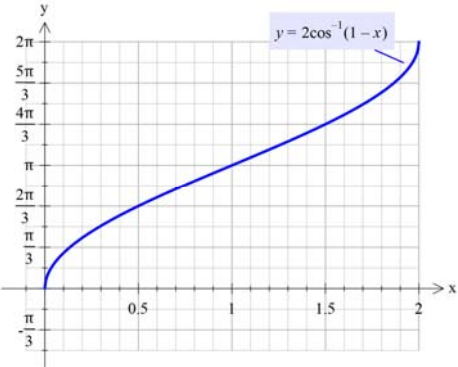
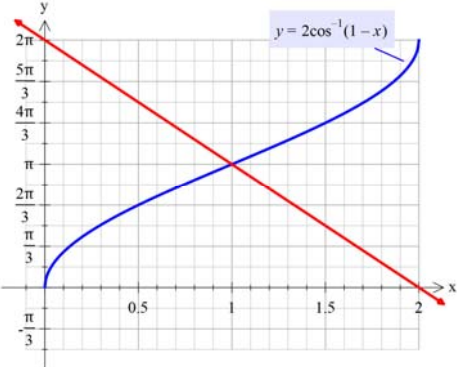
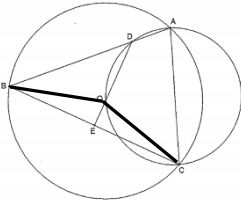
QUESTION 12

Solution		Marks	Comments
12 (a)	$\frac{\cos^2\left(\frac{\pi}{2} - \theta\right) - 2\cos^2\theta}{24} = \frac{\sin^2\theta - 2\cos^2\theta}{24}$ $= \frac{1 - 3\cos^2\theta}{24}$ Now $0 \leq \cos^2\theta \leq 1$ Thus the greatest value of $\frac{\cos^2\left(\frac{\pi}{2} - \theta\right) - 2\cos^2\theta}{24}$ is $\frac{1}{24}$	3	3 marks • Correct solution 2 marks • Simplifies the fraction to a point where only one term is dependent upon θ • Finds the value of θ that gives a maximum 1 mark • Attempts to simplify the expression by using a valid trig identity • Makes a valid attempt to find the value of θ that will give a maximum
12(b) (i)	$\frac{x-5}{x-1} \leq -1$ $x-1 \neq 0$ $x \neq 1$ $x-5 = 1-x$ $2x = 6$ $x = 3$  $1 < x \leq 3$	3	3 marks • Correct graphical solution on number line or algebraic solution, with correct working 2 marks • Bald answer • Identifies the two correct critical points via a correct method • Correct conclusion to their critical points obtained using a correct method 1 mark • Uses a correct method • Acknowledges a problem with the denominator. 0 marks • Solves like a normal equation, with no consideration of the denominator.
12(b) (ii)	$\frac{\cos\alpha - 4}{\cos\alpha} \leq -1$ let $\cos\alpha = u - 1$ $\frac{u-5}{u-1} \leq -1$ $1 < u \leq 3$ $1 < \cos\alpha + 1 \leq 3$ $0 < \cos\alpha \leq 2$ $0 \leq \alpha < \frac{\pi}{2}$	2	2 marks • Correct solution 1 mark • Uses a valid substitution to transform the inequation into part (i) • Establishes the boundary values for the inequation.
12 (c) (i)	$P = 80 + Ae^{kt}$ $\frac{dP}{dt} = Ake^{kt}$ $= k(80 + Ae^{kt} - 80)$ $= k(P - 80)$	1	1 mark • Correct solution
12 (c) (ii)	when $t = 0$, $300 = 80 + Ae^0$ $= 80 + A$ $A = 220$ $t = 20$, $P = 162$ $162 = 80 + 220e^{20k}$ $e^{20k} = \frac{82}{220}$ $20k = \ln\left(\frac{41}{110}\right)$ $k = \frac{1}{20}\ln\left(\frac{41}{110}\right) = -0.0493\dots$ ∴ the population is decreasing at a rate of approximately 5% per year	2	2 marks • Correct solution 1 mark • Establishes the value of A

Solution	Marks	Comments
12 (c) (iii) when $t = 55$, $P = 80 + 220e^{\frac{55k}{110}}$ $= 80 + 220\left(\frac{41}{110}\right)^{\frac{55}{20}}$ $= 94.5796\dots$ $\therefore$ in 2050 the population is predicted to be 95	1	1 mark • Correct solution <i>Note: no penalty for rounding error</i>
12 (d) Let $\angle XBC = \alpha$ $\angle BAC = \angle XBC = \alpha$ (alternate segment theorem) $\angle BCD = \angle XBC = \alpha$ (alternate $\angle$'s = , $XY \parallel CD$) $\angle BCD = \angle BAD = \alpha$ ($\angle$'s in the same segment) $\therefore \angle BAD = \angle BAC$ Thus AB bisects $\angle CAD$	3	3 marks • Correct proof 2 marks • Establishes two or more angles that are equal to $\angle XBC$, or equivalent • Correct solution with poor reasoning 1 mark • Uses a valid circle geometry theorem with correct reasoning
QUESTION 13		
13 (a) $\frac{dA}{dt} = \pi^2 r$ $A = \pi r^2$ $\frac{dr}{dt} = \frac{dA}{dt} \times \frac{dr}{dA}$ $\frac{dA}{dr} = 2\pi r$ $= \pi^2 r \times \frac{1}{2\pi r}$ $= \frac{\pi}{2} \text{ cm/s}$	2	2 marks • Correct solution 1 mark • Uses the chain rule to combine two or more rates into a single expression
13 (b) (i) $\angle NXB = 152^\circ$ & $\angle NXA_{\text{reflex}} = 260^\circ$ $\angle BXS + \angle NXB = 180$ (straight $\angle NXS$) $\angle BXS + 152^\circ = 180^\circ$ $\angle BXS = 28^\circ$ $\angle AXS + \angle NXS = \angle NXA_{\text{reflex}}$ (common $\angle$) $\angle AXS + 180^\circ = 260^\circ$ $\angle AXS = 80^\circ$ $\angle AXB = \angle AXS + \angle BXS$ (common $\angle$) $\angle AXB = 80^\circ + 28^\circ$ $= 108^\circ$	1	1 mark • Correct explanation <i>Note: formal geometrical proof not required, a simple explanation will suffice</i>
13 (b) (ii) $\frac{h}{AX} = \tan 12^\circ$ $AX = \frac{h}{\tan 12^\circ} = h \cot 12^\circ = h \tan 78^\circ$	1	1 mark • Correct answer
13 (b) (iii) $AX = h \tan 78^\circ$, similarly $BX = h \tan 80^\circ$ $AB^2 = AX^2 + BX^2 - 2AX \cdot BX \cdot \cos \angle AXB$ $800^2 = h^2 \tan^2 78^\circ + h^2 \tan^2 80^\circ - 2h^2 \tan 78^\circ \tan 80^\circ \cos 108^\circ$ $h^2 = \frac{800^2}{\tan^2 78^\circ + \tan^2 80^\circ - 2 \tan 78^\circ \tan 80^\circ \cos 108^\circ}$ $h = \frac{800}{\sqrt{\tan^2 78^\circ + \tan^2 80^\circ - 2 \tan 78^\circ \tan 80^\circ \cos 108^\circ}}$ $h = 95.08532019\dots$ $= 95 \text{ metres (to the nearest metre)}$	2	2 marks • Correct solution 1 mark • Uses the cosine rule in an attempt to find the height <i>Note: no penalty for rounding error</i>

Solution	Marks	Comments
13 (c) When $n = 1$; $LHS = \frac{5-4}{5} = \frac{1}{5}$ $RHS = \frac{1}{5^1} = \frac{1}{5}$ $\therefore LHS = RHS$ Hence the result is true for $n = 1$ Assume the result is true for $n = k$ $\text{i.e. } \sum_{r=1}^k \frac{5-4r}{5^r} = \frac{k}{5^k}$ Prove the result is true for $n = k + 1$ $\text{i.e. Prove } \sum_{r=1}^{k+1} \frac{5-4r}{5^r} = \frac{k+1}{5^{k+1}}$ PROOF: $\begin{aligned} \sum_{r=1}^{k+1} \frac{5-4r}{5^r} &= \sum_{r=1}^k \frac{5-4r}{5^r} + \frac{5-4(k+1)}{5^{k+1}} \\ &= \frac{k}{5^k} + \frac{1-4k}{5^{k+1}} \\ &= \frac{5k + 1 - 4k}{5^{k+1}} \\ &= \frac{k+1}{5^{k+1}} \end{aligned}$ Hence the result is true for $n = k + 1$, if it is true for $n = k$ Since the result is true for $n = 1$, then it is true for all positive integers by induction.	3	There are 4 key parts of the induction; 1. Proving the result true for $n = 1$ 2. Clearly stating the assumption and what is to be proven 3. Using the assumption in the proof 4. Correctly proving the required statement 3 marks • Successfully does all of the 4 key parts 2 marks • Successfully does 3 of the 4 key parts 1 mark • Successfully does 2 of the 4 key parts
13 (d) (i) P lies on the normal at Q $(2ap, ap^2) \Rightarrow x + qy = aq^3 + 2aq$ $2ap + ap^2q = aq^3 + 2aq$ $q^3 - p^2q + 2q - 2p = 0$ $q(q^2 - p^2) + 2(q - p) = 0$ $q(q - p)(q + p) + 2(q - p) = 0$ $(q - p)[q(q + p) + 2] = 0$ $\therefore$ as $p \neq q$, $q^2 + pq + 2 = 0$	2	2 marks • Correct solution 1 mark • Substitutes P into the equation for the normal at Q
13 (d) (ii) $m_{QR} = \frac{aq^2 - ar^2}{2aq - 2ar} = \frac{a(q+r)(q-r)}{2a(q-r)} = \frac{q+r}{2}$ $y - aq^2 = \frac{(q+r)}{2}(x - 2aq)$ $2y - 2aq^2 = (q+r)x - 2aq^2 - 2aqr$ $(q+r)x - 2y = 2aqr$	2	2 marks • Correct proof 1 mark • Find the slope of the chord QR
13 (d) (iii) when $x = 0$, $-2y = 2aqr$ $y = -aqr$ $q^2 + pq + 2 = 0$ similarly $r^2 + pr + 2 = 0$ Thus q and r are the roots of the quadratic $t^2 + pt + 2 = 0$ $\alpha\beta = 2$ $qr = 2$ $\therefore$ when $x = 0$, $y = -2a$ i.e. QR always passes through $(0, -2a)$	2	2 marks • Correct proof 1 mark • Shows $qr = 2$ • Identifies $(0, -2a)$ as the y-intercept

QUESTION 14

Solution	Marks	Comments
14 (a) (i) domain: $-1 \leq 1-x \leq 1$ $-2 \leq -x \leq 0$ $0 \leq x \leq 2$ range: $0 \leq \frac{y}{2} \leq \pi$ $0 \leq y \leq 2\pi$	2	2 marks • Correct answer 1 mark • Finds either domain or range
14 (a) (ii) 	1	1 mark • Correct sketch
14 (a) (iii) 	1	1 mark • Correct sketch
14 (a) (iv) $y = 2\cos^{-1}(1-x)$ has rotational symmetry about the point $(1, \pi)$. Thus the area between $y = 2\cos^{-1}(1-x)$ and $y = -\pi x + 2\pi$ is the same from $x = 0$ to $x = 1$ as from $x = 1$ to $x = 2$	1	1 mark • Correct explanation involving symmetry
14 (a) (v) $\int_0^2 2\cos^{-1}(1-x) dx = \frac{1}{2} \times 2 \times 2\pi = 2\pi$	1	1 mark • Correct solution
14 (b) (i) $\angle BOC = 2\angle BAC$ ($\angle$ at centre, twice $\angle$ at circumference on same arc) $\angle EOC = \angle BAC$ (exterior $\angle$, cyclic quad) $\angle BOC = \angle BOE + \angle EOC$ (common $\angle$) $\angle BOE = \angle BOC - \angle EOC$ $= 2\angle BAC - \angle BAC$ $= \angle BAC$ 	3	3 marks • Correct proof 2 marks • Correct proof with poor reasoning • Uses two different valid circle geometry theorems with correct reasoning 1 mark • Uses a valid circle geometry theorem with correct reasoning
14 (b) (ii) $OB = OC$ (= radii) $\angle BOE = \angle EOC$ (proven in (i)) OE is common $\therefore \triangle BOE \equiv \triangle COE$ (SAS) Thus $BE = CE$ (corresponding sides in $\equiv \Delta$'s)	2	2 marks • Correct proof 1 mark • Significant progress • Correct proof with poor reasoning

Solution	Marks	Comments
14 (c) (i) $P(a+b) = (a+b+c)^3 - 6(a+b+c)(a^2+b^2+c^2) + 8(a^3+b^3+c^3)$ If $(a+b-c)$ is a factor then $P(c) = 0$ $P(a+b) = [(a+b)+c]^3 - 6[(a+b)+c][(a+b)^2 - 2ab + c^2] + 8[(a+b)(a^2 - ab + b^2) + c^3]$ $= [(a+b)+c]^3 - 6[(a+b)+c][(a+b)^2 - 2ab + c^2] + 8[(a+b)[(a+b)^2 - 3ab] + c^3]$ $P(c) = [c+c]^3 - 6[c+c][(c^2 - 2ab + c^2) + 8[c(c^2 - 3ab) + c^3]$ $= (2c)^3 - 12c(2c^2 - 2ab) + 8c(2c^3 - 3abc)$ $= 8c^3 - 24c^3 + 24abc + 16c^3 - 24abc$ $= 0$ $\therefore (a+b-c)$ is a factor	2	2 marks • Correct solution 1 mark • Demonstrates knowledge of the factor theorem
14 (c) (ii) If $(a+b-c)$ is a factor then so is $(a+c-b)$ and $(b+c-a)$ $\therefore (a+b+c)^3 - 6(a+b+c)(a^2+b^2+c^2) + 8(a^3+b^3+c^3) = k(a+b-c)(a+c-b)(b+c-a)$ Equating coefficients of a^3 $a^3 - 6a^3 + 8a^3 = -k a^3$ $k = -3$ $\therefore (a+b+c)^3 - 6(a+b+c)(a^2+b^2+c^2) + 8(a^3+b^3+c^3) = -3(a+b-c)(a+c-b)(b+c-a)$	2	2 marks • Correct proof 1 mark • Finds another factor of the expression